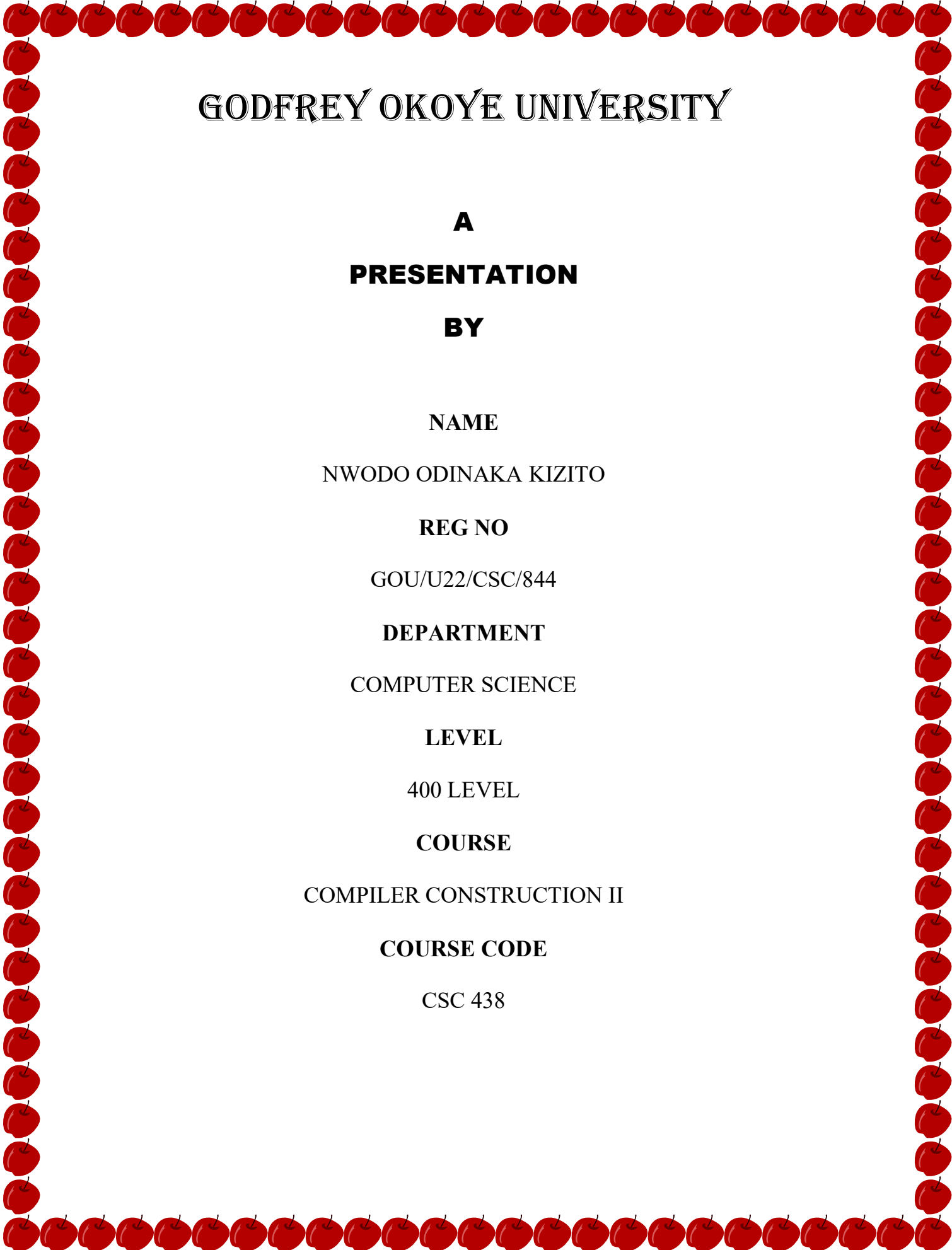




GODFREY OKOYE UNIVERSITY

A PRESENTATION BY

NAME

NWODO ODINAKA KIZITO

REG NO

GOU/U22/CSC/844

DEPARTMENT

COMPUTER SCIENCE

LEVEL

400 LEVEL

COURSE

COMPILER CONSTRUCTION II

COURSE CODE

CSC 438

A) ENTITIES AND ATTRIBUTES

Entities and Attributes

1. Student

- **student_id** (*Primary Key*)
- name
- email
- level

2. Supervisor

- **supervisor_id** (*Primary Key*)
- name
- email
- department

3. Project

- **project_id** (*Primary Key*)
- title
- description

4. Research Area

- **area_id** (*Primary Key*)
- area_name (*e.g., Artificial Intelligence, Cybersecurity*)

CARDINALITIES

1. Supervision Relationship

- A Supervisor can supervise multiple Students
- Each Student is assigned to only one Supervisor
- Cardinality: One-to-Many ($1 : N$) from Supervisor to Student

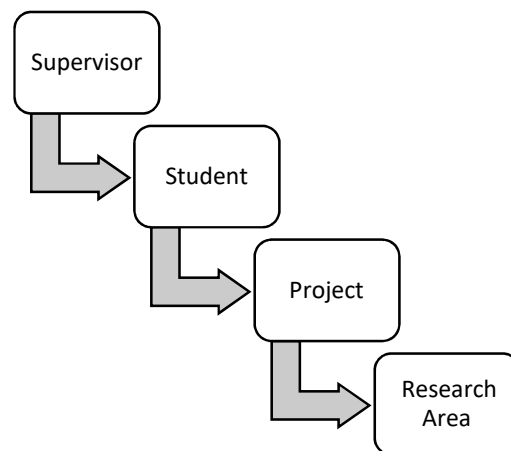
2. Student–Project Relationship

- Each Student works on exactly one Project
- Each Project is assigned to one Student
- Cardinality: One-to-One ($1 : 1$)

3. Project–Research Area Relationship

- Each Project belongs to one Research Area
- A Research Area can contain multiple Projects
- Cardinality: One-to-Many ($1 : N$) from Research Area to Project

ER DIAGRAM



B) RELATIONAL SCHEMA WITH KEYS

Normalized Tables (3NF)

1) Supervisor

```
Supervisor(  
    supervisor_id INT PRIMARY KEY,  
    name VARCHAR(100),  
    email VARCHAR(100) UNIQUE,  
    department VARCHAR(100)  
);
```

2) Student

```
Student(  
    student_id INT PRIMARY KEY,  
    name VARCHAR(100),  
    email VARCHAR(100) UNIQUE,  
    level INT,  
    supervisor_id INT,  
    FOREIGN KEY (supervisor_id)  
    REFERENCES Supervisor(supervisor_id)  
    ON DELETE SET NULL  
    ON UPDATE CASCADE  
);
```

3) Research Area

```
ResearchArea(  
    area_id INT PRIMARY KEY,  
    area_name VARCHAR(100) UNIQUE  
);
```

4) Project

```
Project(  
    project_id INT PRIMARY KEY,  
    title VARCHAR(200),  
    description TEXT,  
    student_id INT UNIQUE,
```

```
area_id INT,  
FOREIGN KEY (student_id) REFERENCES Student(student_id)  
ON DELETE CASCADE,  
FOREIGN KEY (area_id) REFERENCES ResearchArea(area_id)  
ON DELETE SET NULL  
);
```

C) SQL QUERIES

1. LIST ALL STUDENT SUPERVISED BY A PARTICULAR LECTURER

```
SELECT s.student_id, s.name, s.email  
FROM Student s  
JOIN Supervisor sp ON s.supervisor_id = sp.supervisor_id  
WHERE sp.name = 'Dr. John Doe';
```

2. RETRIEVE PROJECTS IN THE “ARTIFICIAL INTELLIGENCE”
RESEARCH AREA

```
SELECT p.project_id, p.title, p.description  
FROM Project p  
JOIN ResearchArea r ON p.area_id = r.area_id  
WHERE r.area_name = 'Artificial Intelligence';
```

3. COUNT NUMBER STUDENT PER SUPERVISOR

```
SELECT sp.supervisor_id, sp.name, COUNT(s.student_id) AS  
total_students  
FROM Supervisor sp  
LEFT JOIN Student s ON sp.supervisor_id = s.supervisor_id  
GROUP BY sp.supervisor_id, sp.name;
```